

EDUCATION

Cornell University

Aug 2026 – present

PhD in Astronomy and Space Sciences

- Studying the landscape of collapsar phenomena with Prof. Anna Ho.

University of Cambridge

Oct 2025 – Jun 2026

MASt in Astrophysics, ranked 1st in cohort

- Thesis: “The effect of the heat-flux buoyancy instability on heat transport in the intracluster medium” (advised by Prof. Henrik Latter and Dr. David Hosking)
- Supported by Trinity College’s Sheepshanks Studentship in Astronomy.

Columbia University

Aug 2021 – May 2025

B.A. in Physics and Mathematics, GPA: 4.03/4.0

- Graduated magna cum laude with departmental honors in physics.

RESEARCH EXPERIENCE

DAMTP Astrophysics Group, University of Cambridge

Oct 2025 – Aug 2026

MASt/Part III Astrophysics Project

- Used analytic and numerical methods to characterize the linear growth of the heat-flux buoyancy instability (HBI)
- Ran 2D and 3D Boussinesq MHD simulations to determine the HBI’s saturation modes across different buoyancy stratifications and magnetic field orientations
- Quantified the heat flux the HBI can feasibly support in the intracluster medium of galaxy clusters.

Department of Astronomy, California Institute of Technology

Jun – Aug 2024

Summer Undergraduate Research Fellow

- Explored effects of dust on tidal disruption event (TDE) emission with the aim of using IR dust echo emission to distinguish between circumnuclear dust geometries and break degeneracies between competing theoretical models of intrinsic emission from TDEs
- Wrote Python scripts to take input high-energy lightcurves and SEDs, propagate them through dust and gas (via iterative radiative transfer calculations in azimuthally symmetric geometries), and calculate “observed” emission.

Department of Physics, Columbia University

May – Dec 2023

Undergraduate Research Assistant

- Investigated the chiral magnetic effect’s (CME) ability to generate large-scale magnetic fields in proto-neutron stars
- Developed an analytic scaling for the maximum magnetic field strength at saturation with the neutronization
- Verified this scaling with MHD simulations in Dedalus.

Black Hole Initiative, Harvard & Smithsonian Center for Astrophysics

May – Dec 2022

NSBP-SAO Event Horizon Telescope Scholar

- Simulated ngEHT polarized imaging pipeline under various array configurations to determine optimal design choices (e.g. mixed polarization bases, rotation between measurements) for imaging certain sources
- Developed Hamiltonian Monte-Carlo forward model of imaging pipeline to estimate parameter covariances
- Developed a polarized M-Ring model for Comrade (ngEHT imaging and analysis library) in Julia
- Produced temperature profiles from GRMHD simulations of jet/accretion flow/black hole (JAB) systems.

PUBLICATIONS

- **N. Nigam**, D. N. Hosking, H. Latter (in prep.)
The effect of the heat-flux buoyancy instability on heat transport in the intracluster medium.
- R. Anantua, ..., **N. Nigam** et al. (Dec 2022)
Emission Modeling in the EHT-ngEHT Age. Galaxies, 11(4).

PRESENTATIONS

- **N. Nigam**, J. J. Somalwar, V. Ravi (Jan 2025)
Simulating the effects of dust on multiwavelength emission from tidal disruption events.
Poster Session 402, American Astronomical Society 245th Meeting, National Harbor, MD.
- **N. Nigam**, V. A. Skoutnev, A. M. Beloborodov (Mar 2024)
Constraining the chiral magnetic effect's ability to generate magnetic fields in proto-neutron stars.
Poster Session, Simons Collaboration on Extreme Electrodynamics of Compact Sources Annual Meeting, New York, NY.
- **N. Nigam**, L. Blackburn (Jan 2022)
Polarization design considerations for the Next Generation Event Horizon Telescope.
Poster Session 471, American Astronomical Society 241st Meeting, Seattle, WA.

AWARDS, HONORS & FELLOWSHIPS

- Sheepshanks Studentship in Astronomy, Trinity College, University of Cambridge Oct 2025 – Jun 2026
- Summer Undergraduate Research Fellow, Caltech Department of Astronomy Jun – Aug 2024
- Feder Family Internship Fund Recipient, Columbia University Center for Career Education May – Aug 2023
- NSBP-SAO Event Horizon Telescope Scholar, Harvard & Smithsonian Center for Astrophysics May – Dec 2022

TEACHING

Columbia University Physics Department

Sep 2024 – May 2025

Recitation Teaching Assistant

- Recitation TA for PHYS1601 (Mechanics and Relativity) and PHYS1602 (Electricity and Magnetism)
- Taught two weekly recitations of 30 students each; wrote weekly problems; facilitated group problem solving.

Columbia University Mathematics Department

Jan – Aug 2023

Undergraduate Teaching Assistant

- Hosted weekly office hours and graded students' assignments [Introduction to Higher Math (MATH2000) in Spring 2023; Linear Algebra (MATH2010) in Summer 2023].

OUTREACH & VOLUNTEERING

Columbia Physics Community Committee

Sep 2024 – May 2025

Undergraduate Representative

- Analyzed, summarized, and distributed results of 2024 department climate survey; prepared 2025 survey
- Brainstormed and launched initiatives to improve departmental inclusion, including a graduate-undergraduate mentorship program
- Attended monthly planning meetings with faculty, postdoctoral scholars, research scientists, and graduate students.

Reading Team Math

Oct 2023 – Dec 2024

Kindergarten Volunteer

- Weekly volunteering with kindergartners at a Harlem after-school program to improve numerical literacy and build confidence with math.

Columbia Spectra & Columbia BlueShift

Nov 2022 – May 2025

Public Relations & Outreach Chair

- Ran social media, made graphics and flyers, and organized and publicized events for undergraduate outreach clubs in physics and astronomy.

WORKSHOPS

- Perimeter Institute, PSI Start Summer School on Topics in Theoretical Physics May – Jul 2024
- New York University Artificial Intelligence School Jun 2023

SKILLS & INTERESTS

Software: Python, Julia, C, Octave, Mathematica, HTML/CSS, LaTeX, Git

Languages: English, German, Hindi, Mandarin, French

Interests: Music (classical, jazz, rock), dance, tennis, running